

Lead Climbing

Harness

Anatomy of a Harness

Gear Loops

- *Not* safety rated tie in points.
- You know you waist loop in on correctly when the gear loops are facing down.

Waist Loop

- Safety critical portion of harness.
- Above hip bones.
- Shouldn't be able to slip it off:
 - Place flat hand through waist loop then make a fist and attempt to pull it out.
 - Should be able to breath.

Leg Loops

- Should be comfortable.
- Not safety critical.
- Adjust leg loops where you want them, then adjust strap in back:
 - The higher up on your leg the leg loops are, the more upright you'll be in your harness when sitting in it.
- Back strap is sometimes removable:
 - This is so you can go to the bathroom.

Tie-in Points

- Rope goes through both of them.
- Should always be through both:
 - What if only the bottom is used:
 - You won't be directly connected to the waist loop.
 - You'll flip up-side-down when you fall.
 - What if only the top is used:
 - It will affect how you sit and take falls.
 - Is less safe.
- Bottom up or top down?
 - Doesn't matter.
 - Most go bottom up.
 - Some say it's dangerous because if you only go through the bottom one, you aren't tied directly into the waist loop.

Belay loop

- Goes through both tie-in points.
- Used by belayer, not the climber:
 - There are some exceptions, like auto belays.

Check Harness

- Tie-in points and belay loop should be on front.

- Ensure gear loops are protruding from the bottom of the waist loop.
- Straps connecting bottom tie-in point to leg loops should not be twisted.
- Strap on back of leg loops should be on the top.

Figure 8

Versions of the Figure 8

A figure 8 knot can have the load strand on the inside or outside.

The location of the load strand matters. It's not a safety issue, but having the load strand on the outside will make it harder to un-tie.

Tail

Correct Amount of Tail

A fists worth, unless you have small fists.

Note that when taking the lead test at Movement, they want you to call out that you have 6 inches of tail.

Dealing with the Tail

- Tie a knot:
 - This knot is often called a stopper knot or a safety knot:
 - This knot adds *no* safety and the figure 8 need not be more safe.
 - The generic name of this knot is... well, there are many:
 - Barrel/Trippl barrel knot
 - Overhand/Trippl overhand knot
 - If you have very little tail, you may only be able to tie a barrel knot.
 - If you have more tail, you should tie a trippl barrel knot.
- Yosemite finish:
 - Won't teach you this, if you do it wrong, you'll die.

Dressed

Tying a dressed knot makes it easier to un-tie, also, it looks better. You'll know it's dressed because all strands will be parallel.

Ropes

There are two parts of ropes, the core and the sheath.

Don't step on ropes, dirty ropes don't last as long and wear out the carabiners in quick draws.

Climbing ropes are dynamic. This means that they stretch. Semi-dynamic ropes also exist. They are for use with ascending devices.

Rope Core

The core of ropes is normally made out of Dyneema. Dyneema is a branded name for ultra-high-molecular-weight polyethylene (UHMWPE) fiber.

Dyneema can't be died, so if it's white, it's probably Dyneema.

If you ever see white showing through in your rope, don't climb on it.

Rope Sheath

This is the layer that wraps the core of the rope. It's there to protect the Dyneema core.

Checking Ropes

Ropes should be checked prior to climbing on them. This includes:

- Looking for white code showing through the sheath.
- Feeling for narrow portions.
- Running the pinch test on narrow portions of rope.

Preparing the Rope

- Decide where the rope should be placed:
 - There is no hard and fast rule.
 - When climbing outside obstacles at the base of the crag are going to be the deciding factor.
 - I tend to flake the rope under or to the right of the bolts.
 - If the bolts are to the left of the start of the climb, I'll flake the rope under the bolts.
 - If the bolts are to the right or in line with the start of the climb, I'll flake the rope to the right of the bolts.
- Tie stopper knot (triple barrel knot).
- Flake the rope:
 - As you flake the rope, check it for damage.
- There should be plenty of slack between the belay device and the climber at the start:
 - This is so you can spot the climber while they climb to the first bolt without short roping them.

Belay Devices

There are 4 categories of belay devices:

- Tubular Belay Device (Tubers):
 - Often called ATCs, but that's like calling all soda Coke.
- Assisted Tubular Belay Device (Assisted Tubers):
 - These can lock.
 - This reduces the amount of effort needed when holding the brake.
 - You still have to hold the brake.
 - You must unlock them prior to feeding rope.
- Cam-assisted belay devices:
 - The Petzl GRIGRI is the most popular version of this.
- Other:
 - There's some other thing that locks only when the rope goes through it at some speed.

Belay Carabiner

There are many variants of belay carabiners. They all lock, many have a gate to maintain the orientation of the carabiner.

Belayer Check

Ensure:

- Belay carabiner is:
 - Through the belay loop.

- Locked.
- The tail exits the bottom of the belay device.
- The load strand leads to the climber.
- Lever is on the left (if it's a GRIGRI).

System Check

Many call it a buddy check, I don't like this because you can check yourself and your buddy, but be tied into different ropes.

Both belayer and climber must check:

- Themselves.
- Their climbing partner.
- That they are connected to the same rope.
- A triple barrel knot is tied in the belayer's end of the rope.
- The rope is flaked.

Extending Quick Draws

When connecting multiple quick draws together in order to extend them, the bolt carabiner from the second quick draw should be removed.

The order will be: carabiner, dog bone, carabiner, dog bone, carabiner.

Weight Differences

You can use tools to add drag to the rope, tie into an anchor in the ground, or into a weighted bag.

Clipping

First some terminology:

- Quick Draws:
 - The things you clip into on the wall.
- Dog bone:
 - The piece of textile that connects the two carabiners.
- Back clip:
 - Clipping into a carabiner such that the end of the rope that leads to you is facing into the wall:
 - Doing this can lead to the rope coming out of the carabiner.
- Z clip:
 - Using the portion of the rope that's below the last quick draw when clipping into the next quick draw.
 - Causes you to take longer falls.
 - Increases rope drag.

How to Clip into Quick Draw

- The carabiner with the retainer on it is the side you clip the rope into.
- The orientation of the gate should flip with each quick draw placed.
- There are many ways of clipping the rope into the carabiner.
- Know multiple, be able to clip with both hands, regardless of what side the gate is on.
- Don't get your fingers in the gate.

- Clip when the quick draw is between your hips and your chest:
 - Attempt to clip close to waist.
 - If clipping outside of this range, let your belayer know.

Stick Clips

They are tools used to attach quick draws to bolts in the wall from the ground.

Stepping Behind the Rope

Don't let your leg be between the wall and the rope. If it is, and you fall, your leg will get trapped when the rope tightens and flip you up-side-down.

Practice Climb

- Take a fall from the top rope.
- Remember to continue clipping from where you fell.
- Clip into both mousies/carabiners at the top.
- Don't step behind the rope:
 - This will cause you to flip upside down when falling.

Belaying

- Spot the climber while they climb to the first bolt.
- When they're about to clip into the first quick draw:
 - Adjust the amount of slack so there is no slack in the system.
 - Move all the way to the side.
 - Keep the rope close to the wall so the climber can step over it easily.
 - Remain close to the wall.
- Stay close to the wall, and minimize the amount of slack, until the climber gets to about the third bolt:
 - As the climber approaches the bolt they've clipped into, *take out* slack.
 - As the climber climbs above the last bolt they've clipped into, *feed out* slack.
- Once the climber has reached the third bolt:
 - Center yourself with the first bolt.
 - Take a step back.
 - More slack may exist in the system.
- Don't short rope your climber:
 - Ideally ever.
 - Doing so can pull your climber off the wall.
 - Having some slack in the system helps with soft catches:
 - Soft catches prevent ankles from getting broken.
- Provide soft catches to your climber:
 - Only time you shouldn't, is when doing so would cause them to deck.
- As you belay, get in the habit of flicking the break side of the rope:
 - This should be done so you can ensure that the rope is not pinned under anything, including your foot, without having to look away from your climber.
 - Belayers standing on the rope is a common reason for short roping a climber.

Feeding Out Slack

There are two common methods of feeding slack from a Petzl GRIGRI, flossing and depression of the cam.

Soft Catches

A soft catch is when you stop the fall of the climber in a less abrupt way.

Soft catches are important not because of the vertical portion of the fall will hurt the climber, it won't. They're important because they reduce the horizontal component. This reduces the speed the climber approaches the wall at. Doing this prevents ankles from getting broken.

Hard catches are also harder on the equipment. This doesn't mean the equipment will fail, but it won't last as long, and it's expensive.

Soft Catch Methods

There are just about as many ways to provide soft catches as there are hair cuts.

- Allowing rope to slip through belay device:
 - Works best on tubers.
 - Could work with some assisted tubers, maybe?
- Running towards the wall:
 - Tricky.
 - Prone to tripping/falling.
 - Difficult to be consistent.
 - Doesn't provide *as* soft as a catch.
 - If you're far from the wall, you *have* to do this:
 - If you don't, you'll just get pulled into the wall regardless.
- Jumppping:
 - When you see your climber falling, bend you legs, then jump.
 - Stand close to the wall.
 - Helps to have one leg forward and be loose rather than stiff.
 - If there is a lot of rope drag, or you're significantly heavier than your climber:
 - Bend legs in anticipation of fall.
 - Do this one.
- Some combination of the above.
- Do nothing:
 - This could still result in a soft catch if the belayer is lighter than the climber.
- Take slack out of system:
 - Running away from the wall, hitting the ground, etc.
 - This normally *doesn't* result in a soft catch.

Do Soft Catches Require Extra Slack?

Soft catches *don't* require extra slack in the system. Having *some* slack in the system *can* assist with soft catching. This is because of the dynamic nature of climbing ropes.

When there's more rope between the belayer and climber, the rope will expand more and result in lower forces to the climber. If the climber is already high up, there is *no* need to add extra slack. When the climber is low down, *extra slack* could result in a **ground fall!**

Have *no* slack makes timing the soft catch harder.

So What's the Correct Amount of Slack?

One, *at most* two, elbows worth of slack.

When the climber is high up, far from the last bolt, or you're further away from the wall, less is needed. For example, if the climber is far past the last bolt, you may want to take in slack while *also* providing a soft catch. Note that you *must* be very fast when taking out slack.

Adding slack can be helpful if the climber has just traversed a roof section. At that point, you would like the climber to swing past the current section of wall, and into the empty space below it. This prevents your climber from losing teeth.

Post Fall Pull Up

Once your climber has fallen, they will want to get back to where they were before. This can be done by them re-climbing the route, or by pulling back up.

If the route is hard, the climber won't want to re-climb it.

To allow your climber to pull back up:

- Take all slack out of the system.
- Put all of your weight in the harness.
- If you're lighter than the climber, you may need to time a jump with when the climber is attempting to pull back up.